

Exercise Solutions: Arrays

1. Read and Display Array Elements

Solution

```
import java.util.Scanner;

public class DisplayArray {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[] numbers = new int[5];

        System.out.print("Input 5 integers: ");
        for (int index = 0; index < numbers.length; index++) {
            numbers[index] = input.nextInt();
        }

        System.out.println("The array elements are:");
        for (int index = 0; index < numbers.length; index++) {
            System.out.print(numbers[index]);
            if (index < numbers.length - 1) {
                System.out.print(" ");
            }
        }
        System.out.println();
        input.close();
    }
}
```

Sample Output

```
Input 5 integers: 12 7 25 4 18
The array elements are:
12 7 25 4 18
```

2. Find the Sum and Average of Array Elements

Solution

```
public class ArraySumAndAverage {  
    public static void main(String[] args) {  
        int[] numbers = {10, 20, 30, 40, 50};  
        int sum = 0;  
  
        for (int number : numbers) {  
            sum += number;  
        }  
  
        double average = (double) sum / numbers.length;  
        System.out.println("Sum: " + sum);  
        System.out.println("Average: " + average);  
    }  
}
```

Sample Output

```
Sum: 150  
Average: 30.0
```

3. Find the Largest and Smallest Array Elements

Solution

```
public class ArrayLargestAndSmallest {
    public static void main(String[] args) {
        int[] numbers = {45, 12, 78, 3, 56};
        int largest = numbers[0];
        int smallest = numbers[0];

        for (int index = 1; index < numbers.length; index++) {
            if (numbers[index] > largest) {
                largest = numbers[index];
            }
            if (numbers[index] < smallest) {
                smallest = numbers[index];
            }
        }

        System.out.println("Largest element: " + largest);
        System.out.println("Smallest element: " + smallest);
    }
}
```

Sample Output

```
Largest element: 78
Smallest element: 3
```

4. Search for an Element in an Array

Solution

```
import java.util.Scanner;

public class SearchArray {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[] numbers = {8, 15, 22, 31, 47};

        System.out.print("Search value: ");
        int searchValue = input.nextInt();
        int foundIndex = -1;

        for (int index = 0; index < numbers.length; index++) {
            if (numbers[index] == searchValue) {
                foundIndex = index;
                break;
            }
        }

        if (foundIndex >= 0) {
            System.out.println(searchValue + " was found at index " + foundIndex);
        } else {
            System.out.println(searchValue + " was not found in the array");
        }
        input.close();
    }
}
```

Sample Output

```
Search value: 22
22 was found at index 2
```

5. Reverse an Array

Solution

```
public class ReverseArray {  
    public static void main(String[] args) {  
        int[] numbers = {1, 2, 3, 4, 5};  
  
        System.out.println("The reversed array is:");  
        for (int index = numbers.length - 1; index >= 0; index--) {  
            System.out.print(numbers[index]);  
            if (index > 0) {  
                System.out.print(" ");  
            }  
        }  
        System.out.println();  
    }  
}
```

Sample Output

```
The reversed array is:  
5 4 3 2 1
```